

YAKUTSK, Russian Federation

WMO: 249590

Lat: 62.017N

Lon: 129.717E

Elev: 101

StdP: 100.12

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.9	-46.2	-50.1	0.0	-47.7	-48.5	0.0	-46.2	4.8	-32.3	4.3	-32.8	0.7	340	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	30.5	19.0	28.4	18.2	26.5	17.4	20.6	27.6	19.4	26.2	18.3	24.7	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.1	23.6	16.8	12.1	23.1	15.6	11.2	21.7	60.0	27.6	56.1	26.1	52.3	24.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.8	-48.4	33.2	2.5	1.6	-50.2	34.4	-51.6	35.3	-53.0	36.2	-54.8	37.3		
(5)			-48.4	22.5	2.5	1.3	-50.2	23.4	-51.6	24.1	-53.0	24.9	-54.8	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.1	-37.5	-32.9	-18.9	-3.8	7.6	16.4	19.2	15.5	6.2	-7.1	-26.2	-37.0	(6)
(7)		DBStd	21.40	5.45	5.66	7.21	5.76	4.34	4.11	3.78	3.63	4.18	6.54	7.97	5.72	(7)
(8)		HDD10.0	7288	1472	1202	894	414	101	4	1	3	126	529	1085	1457	(8)
(9)		HDD18.3	9738	1730	1435	1153	664	334	84	36	102	363	787	1335	1716	(9)
(10)		CDD10.0	693	0	0	0	0	25	195	286	174	13	0	0	0	(10)
(11)		CDD18.3	103	0	0	0	0	0	25	63	15	0	0	0	0	(11)
(12)		CDH23.3	1304	0	0	0	0	8	371	734	189	2	0	0	0	(12)
(13)		CDH26.7	389	0	0	0	0	1	99	246	43	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.1	1.1	1.7	2.3	2.8	2.6	2.2	2.3	2.0	1.4	1.1	(14)	
(15)	Precipitation	PrecAvg	234	9	7	6	9	19	32	40	38	29	19	16	10	(15)
(16)		PrecMax	326	22	22	21	24	58	87	100	151	86	49	42	21	(16)
(17)		PrecMin	134	3	1	0	0	1	3	0	3	3	4	4	2	(17)
(18)		PrecStd	46	5	4	4	6	14	19	23	27	18	10	8	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.3	-11.2	3.3	13.5	24.6	31.5	33.6	30.4	21.5	7.6	-2.0	-18.1	(19)
(20)			MCWB	-19.6	-11.8	-0.5	5.9	13.2	18.2	20.4	19.5	13.6	4.1	-2.8	-18.4	(20)
(21)		2%	DB	-24.5	-17.9	-1.0	10.0	21.2	29.4	31.3	27.5	18.1	4.8	-7.1	-23.3	(21)
(22)			MCWB	-24.7	-18.3	-3.7	3.9	11.5	17.6	19.6	18.5	11.4	1.8	-7.9	-23.5	(22)
(23)		5%	DB	-27.2	-21.7	-3.8	7.3	18.7	27.4	29.5	25.3	15.7	2.7	-11.2	-26.3	(23)
(24)			MCWB	-27.3	-22.0	-5.9	2.2	9.9	16.7	18.9	17.0	10.5	0.6	-11.7	-26.5	(24)
(25)		10%	DB	-29.7	-24.8	-6.9	5.0	16.2	25.4	27.5	23.2	13.5	1.0	-14.9	-29.0	(25)
(26)			MCWB	-29.8	-25.0	-8.5	0.9	8.7	15.7	18.5	16.1	9.3	-0.7	-15.3	-29.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.6	-11.8	-0.5	6.4	13.8	20.6	22.7	21.1	14.3	4.6	-2.7	-18.6	(27)
(28)			MCDWB	-19.4	-11.3	3.1	12.8	22.8	28.6	29.9	27.4	19.1	7.3	-2.1	-18.4	(28)
(29)		2%	WB	-24.7	-18.4	-3.6	4.2	12.0	18.7	21.2	19.5	12.8	2.3	-7.7	-23.6	(29)
(30)			MCDWB	-24.5	-18.1	-1.0	9.5	19.6	26.7	28.4	25.7	17.0	4.2	-7.1	-23.4	(30)
(31)		5%	WB	-27.4	-22.0	-5.8	2.5	10.5	17.6	20.2	18.1	11.3	0.7	-11.7	-26.5	(31)
(32)			MCDWB	-27.2	-21.7	-3.8	6.8	17.3	25.3	27.2	24.0	14.7	2.4	-11.2	-26.4	(32)
(33)		10%	WB	-29.8	-24.9	-8.5	1.1	9.3	16.5	19.2	16.8	9.8	-0.6	-15.3	-29.1	(33)
(34)			MCDWB	-29.7	-24.7	-7.0	4.7	15.3	23.9	25.9	22.0	13.0	1.0	-14.9	-29.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	8.5	13.4	12.2	11.3	12.1	11.5	11.2	9.6	7.2	6.8	4.9	(35)
(36)			MCDBR	8.9	11.9	14.8	12.6	15.6	16.0	15.5	15.3	13.8	8.3	9.5	8.3	(36)
(37)			MCWBR	8.8	11.5	12.5	7.7	8.0	7.0	6.8	7.6	8.4	6.1	9.2	8.1	(37)
(38)		5% WB	MCDBR	8.9	11.9	14.6	12.0	13.6	13.9	13.0	13.4	11.7	7.9	9.5	8.3	(38)
(39)			MCWBR	8.8	11.6	12.3	7.7	7.3	6.6	6.4	7.2	7.7	6.2	9.1	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.213	0.257	0.299	0.357	0.425	0.419	0.495	0.468	0.362	0.287	0.219	0.183	(40)	
(41)		taud	1.972	2.082	2.075	2.075	2.122	2.253	2.023	2.056	2.322	2.299	1.996	1.890	(41)	
(42)		Ebn at Noon	467	699	796	818	788	804	718	698	732	668	442	305	(42)	
(43)		Edh at Noon	38	71	104	130	136	122	150	133	84	57	35	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.36	1.36	3.13	5.05	5.24	5.96	5.44	4.18	2.56	1.13	0.50	0.17	(44)	
(45)		RadStd	0.02	0.05	0.11	0.19	0.40	0.36	0.32	0.31	0.28	0.12	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) +0.92	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) -306	(k) -328	(l) N/A	(m) N/A	(46)
(47)	Regional (7 neighbors)	+0.71	+1.44	+1.31	N/A	N/A	N/A	-236	-255	N/A	N/A	(47)

Nomenclature: See separate page